Software Project Lab – 1 Midterm Presentation

Project Title: LogWarden (Log-Based Intrusion Detection System)

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Introduction

Why Analyze Server Logs?

Project Description – “What?”

Tool Development

Develop an offline, command-line Log-Based Intrusion Detection Tool that analyzes SSH, web and outbound logs to identify suspicious activity.

Project Description – “How?”

Log Processing

Read SSH, web and outbound logs, parse source formats, and normalize records.

Alerts & Evidence

Generate severity-based alerts with supporting event evidence.

Detection Engine

Apply rule-based detection using counters and sliding time windows.

What I have Implemented So Far…

Log File Handling

- Reads SSH authentication, web access, and outbound network/application logs
- Parses records from each source-specific log format
- Processes multiple input log files in one run
- Displays parsed events in the terminal

What I have Implemented So Far…

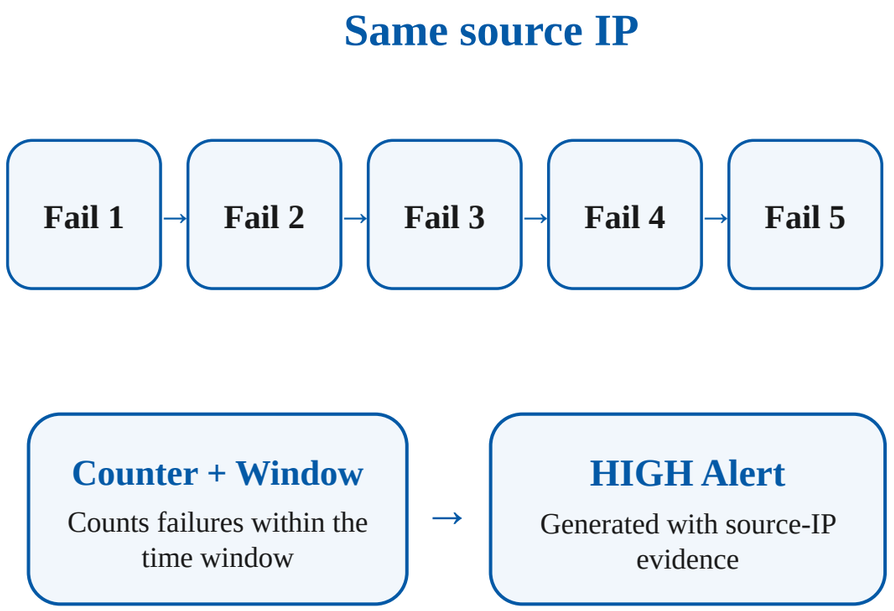
Common Event Normalization

- Converts parsed records into a common event representation
- Captures fields such as type, timestamp, source IP, username, method, target and status
- Displays parsed SSH and web events
- Reports parsing totals by event type

What I have Implemented So Far…

SSH Brute-Force Detection

- Counts failed SSH login attempts for each source IP
- Uses a counter and a sliding time window
- Generates a HIGH-severity alert when the rule is triggered
- Tests cover separate IPs, events outside the time window, and duplicate suppression



Observed test: 5 failures per IP produced HIGH alerts

What I have Implemented So Far…

Successful Login After Repeated Failures

- Tracks failed SSH logins from the same source IP
- Checks whether a successful login follows repeated failures
- Uses a sliding time window and counter
- Generates a CRITICAL alert with supporting event evidence



Evidence

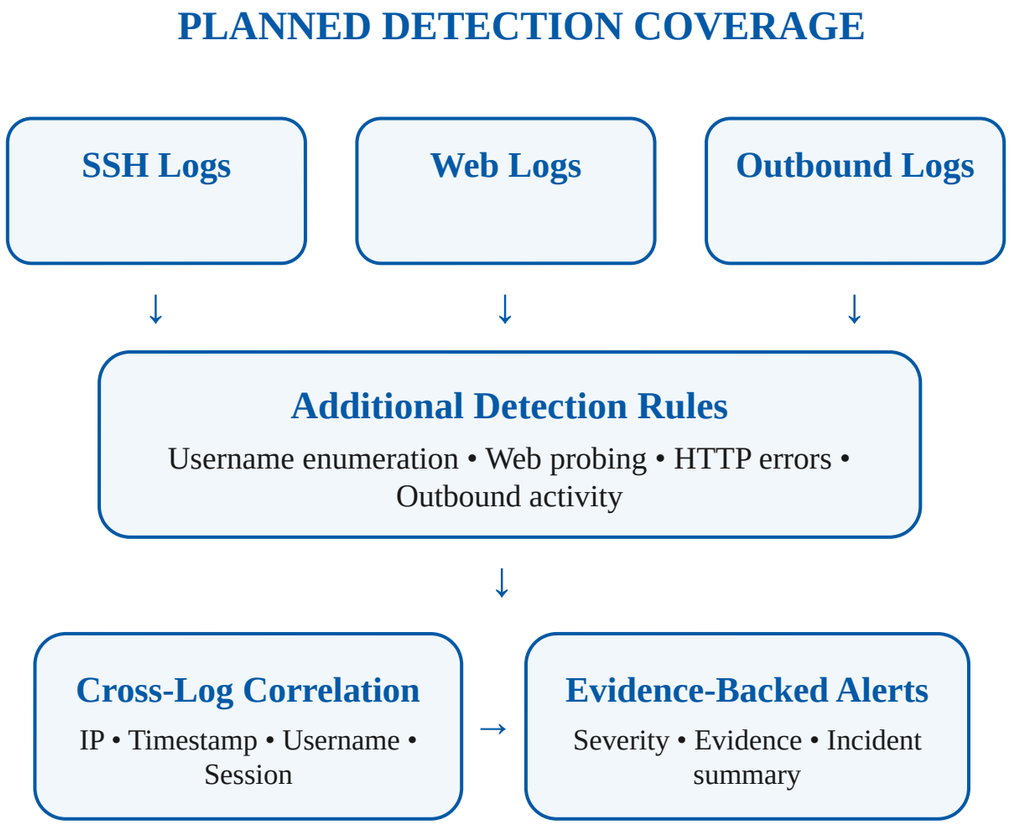
5 failed logins + 1 successful login = 6 evidence events

Test output: 1 CRITICAL alert generated

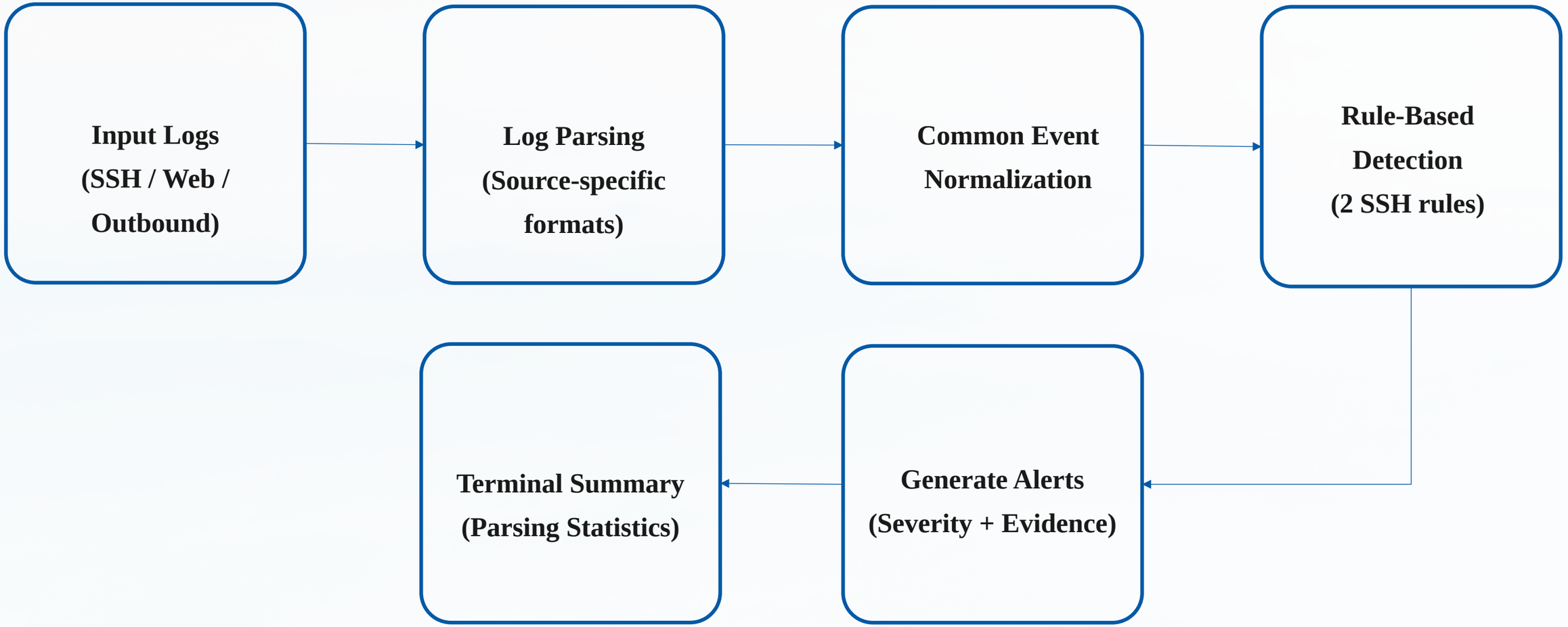
Future Plan

Planned Enhancements

- SSH username enumeration detection
- Web resource probing from the same source IP
- Repeated HTTP errors from the same source IP
- Suspicious outbound activity detection
- Cross-log correlation across related events



Work Flow


Input Logs
(SSH / Web /
Outbound)
Log Parsing
(Source-specific
formats)
Common Event
Normalization
Rule-Based
Detection
(2 SSH rules)
Generate Alerts
(Severity + Evidence)
Terminal Summary
(Parsing Statistics)

Project Screenshots

```
mehenaz_maisha@DESKTOP-1JTEK68:/mnt/e/SPL-1/LogWarden$ g++ Src/parser.cpp Src/Main.cpp
mehenaz_maisha@DESKTOP-1JTEK68:/mnt/e/SPL-1/LogWarden$ ./a.out Logs/LBIDS_combined_access_ssh_outbound.log Logs/comb
ined_ssh_web_intrusion_log.log Logs/student-access-SPL1.log Logs/Synthetic_Log.log

Parsing: Logs/LBIDS_combined_access_ssh_outbound.log

========== Parsed Event 1 ==========
Type: SSH
Timestamp: Sep 19 01:44:25
Source IP: 198.51.100.200
Username: root
Method:
Target:
Status: -1
Port: 32262
Authentication result: FAILED
Response size: -1
Raw log: Sep 19 01:44:25 webserver sshd[33809]: Failed password for root from 198.51.100.200 port 32262 ssh2
```

Parsed SSH Authentication Event

Project Screenshots

```
========== Parsed Event 28 ==========
Type: WEB
Timestamp: 16/Sep/2026:00:05:27 +0600
Source IP: 103.181.104.74
Username:
Method: GET
Target: /bn/408b7c8ad06e4d9954fa2d948a01f508
Status: 302
Port: -1
Authentication result: UNKNOWN
Response size: 5
Raw log: 103.181.104.74 - - [16/Sep/2026:00:05:27 +0600] "GET /bn/408b7c8ad06e4d9954fa2d948a01f508 HTTP/1.1" 302 5 "
-" "Mozilla/5.0 (Linux; Android 10; K) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Mobile Safari/537.36"
```

Parsed Web Access Event

Project Screenshots

```
mehenaz_maisha@DESKTOP-1JTEK68:/mnt/e/SPL-1/LogWarden$ ./a.out

========== TEST A: Separate IPs ==========
Input events: 10
Alerts generated: 2
Alert 1: IP = 198.51.100.200, failures = 5, severity = HIGH
Alert 2: IP = 192.0.2.222, failures = 5, severity = HIGH

========== TEST B: Outside time window ==========
Input events: 5
Alerts generated: 0

========== TEST C: Duplicate suppression ==========
Input events: 7
Alerts generated: 1
Alert 1: IP = 203.0.113.50, failures = 5, severity = HIGH
```

SSH Brute-Force Rule Test

Project Screenshots

```
mehenaz_maisha@DESKTOP-1JTEK68:/mnt/e/SPL-1/LogWarden$ ./a.out
Total test events: 6
Alerts generated: 1

Alert 1
Rule ID: R2
Rule Name: Successful Login After Repeated Failures
Severity: CRITICAL
Source IP: 192.0.2.10
Failed login count: 5
First event: Sep 19 01:00:00
Successful login: Sep 19 01:01:40
Evidence count: 6
```

Successful Login After Failures

Project Screenshots

```
Port: -1
Authentication result: UNKNOWN
Response size: 0
Raw log: 103.181.104.74 - - [16/Sep/2026:00:05:29 +0600] "GET /js/landingPage2.js?v=15 HTTP/1.1" 304 0 "https://www.
xyz.com/bn/408b7c8ad06e4d9954fa2d948a01f508" "Mozilla/5.0 (Linux; Android 10; K) AppleWebKit/537.36 (KHTML, like Gec
ko) Chrome/153.0.0.0 Mobile Safari/537.36"

Parsing: Logs/combined_ssh_web_intrusion_log.log

Parsing: Logs/student-access-SPL1.log

Parsing: Logs/Synthetic_Log.log


========== PARSING SUMMARY ==========
Total files provided: 4
Total lines read: 142256
Successfully parsed: 142234
SSH events: 116
Web events: 142090
Outbound events: 28
Unrecognized lines: 1
Events displayed: 30
```

Parsing Summary

Thank you